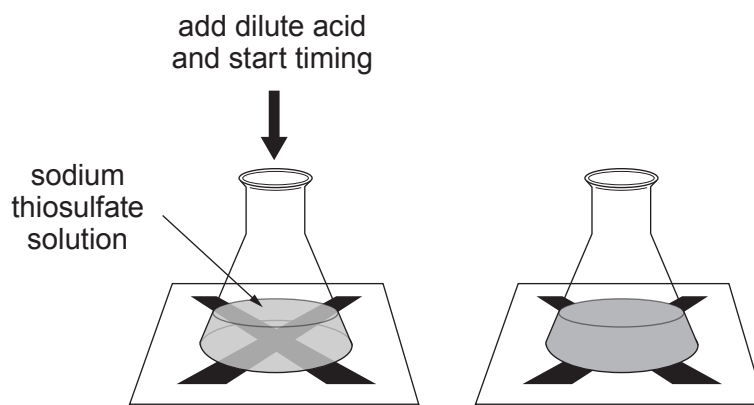


4. Sodium thiosulfate solution reacts with dilute hydrochloric acid to form a precipitate. The precipitate causes the solution to go cloudy.

The rate of the reaction can be measured by placing a cross beneath the flask and measuring the time taken for the cross to disappear.



Gareth and Sion studied the effect of sodium thiosulfate concentration by carrying out the reaction with thiosulfate of five different concentrations. They tested each concentration three times.

Their results are shown in the table below.

Concentration of sodium thiosulfate (g/dm ³)	Time 1 (s)	Time 2 (s)	Time 3 (s)	Mean time (s)
0.2	114	113	112	113
0.4	74	70	72	72
0.6	40	38	57	39
0.8	21	23	22	22
1.0	14	16	15	15

- (a) When calculating the mean times, they ignored one of the values recorded.

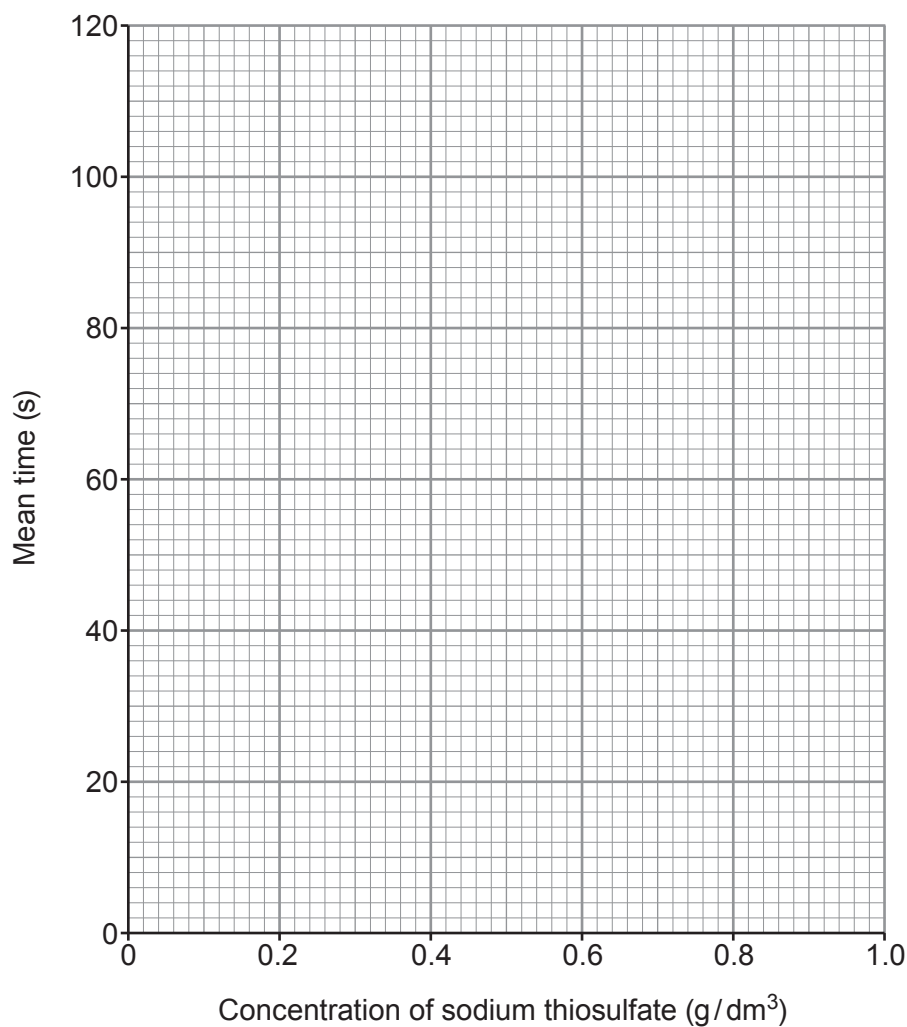
Circle this value in the table.

[1]



- (b) Plot the mean time against sodium thiosulfate concentration on the grid below. Draw a suitable line.

[3]



- (c) Explain the results in terms of particle theory.

[3]

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- (d) A second group of students carried out the same experiment with sodium thiosulfate solution at a higher temperature.

On the grid above, sketch a line to show the results you would expect them to record.

[1]



- (e) The stock solution of sodium thiosulfate used in both experiments was made by dissolving 1.0 g of the solid in 1 dm³ of water.

Calculate the number of moles of sodium thiosulfate (Na₂S₂O₃) in 1.0 g.

Give your answer to **two** significant figures.

[2]

$$A_r(\text{Na}) = 23$$

$$A_r(\text{S}) = 32$$

$$A_r(\text{O}) = 16$$

Number of moles = mol

10

